$$\frac{du}{dv} = \frac{\left(\frac{du}{dx}\right)}{\left(\frac{dv}{dx}\right)}$$

$$= \frac{\left[\frac{2x}{(1+x^2)^2}\right]}{2x}$$

$$= \frac{1}{(1+x^2)^2}$$

Exercise 5.8

1. Find $\frac{dy}{dx}$ of the following functions
   (i) $x = ct,\ y = \frac{c}{t}$
   (ii) $x = \log t,\ y = \sin t$
   (iii) $x = a\cos^3\theta,\ y = a\sin^3\theta$
   (iv) $x = a(\theta - \sin\theta),\ y = a(1 - \cos\theta)$
2. Differentiate $\sin^3 x$ with respect to $\cos^3 x$
3. Differentiate $\sin^2 x$ with respect to x^2

5.3.7 Successive differentiation

The process of differentiating the same function again and again is called successive differentiation.

(i) The derivative of y with respect to x is called the first order derivative and is denoted by $\frac{dy}{dx}$ (or) y_1 (or) $f'(x)$

(ii) If $f'(x)$ is differentiable, then the derivative of $f'(x)$ with respect to x is called the second order derivative and is denoted by $\frac{d^2y}{dx^2}$ (or) y_2 (or) $f''(x)$

(iii) Further $\frac{d^ny}{dx^n}$ (or) y_n (or) $f^{(n)}(x)$ is called n^{th} order derivative of the function $y = f(x)$

Remarks

(i) If $y = f(x)$, then

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right).$$

(ii) If $x = f(t)$ and $y = g(t)$, then

$$\frac{d^2y}{dx^2} = \frac{d}{dt}\left(\frac{dy}{dx}\right)\frac{dt}{dx}$$

$$= \frac{d}{dt}\left\{\frac{g'(t)}{f'(t)}\right\}\cdot\frac{dt}{dx}$$

Example 5.46

Find the second order derivative of the following functions with respect to x,

(i) $3\cos x + 4\sin x$
(ii) $x = at^2,\quad y = 2at$
(iii) $x\sin x$

Solution

(i) Let $y = 3\cos x + 4\sin x$

$y_1 = -3\sin x + 4\cos x$

$y_2 = -3\cos x - 4\sin x$

$y_2 = -(3\cos x + 4\sin x)$

$y_2 = -y$ (or) $y_2 + y = 0$

(ii) $x = at^2,\quad y = 2at$

$\frac{dx}{dt} = 2at \qquad \frac{dy}{dt} = 2a$

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)}$$

$$= \frac{1}{t}$$

Now, $$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$

$$= \frac{d}{dt}\left(\frac{dy}{dx}\right)\frac{dt}{dx}$$

$$= \frac{d}{dt}\left(\frac{1}{t}\right)\cdot\frac{1}{2at}$$

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